

GENE 07959

Small mobilizable multi-purpose cloning vectors derived from the *Escherichia coli* plasmids pK18 and pK19: selection of defined deletions in the chromosome of *Corynebacterium glutamicum*

(Broad-host-range conjugation; kanamycin resistance; sequencing vector; *lacZ*; gene disruption and replacement; *sacB* gene; coryneform bacteria)

Andreas Schäfer^a, Andreas Tauch^a, Wolfgang Jäger^a, Jörn Kalinowski^a, Georg Thierbach^b and Alfred Pühler^a

^aDepartment of Genetics, Faculty of Biology, University of Bielefeld, 33501 Bielefeld, Germany; and ^bDegussa AG/ASTA Medica AG, Kantstr. 2, 33790 Halle, FRG. Tel. (49-5201) 511

Received by M. Bagdasarian: 16 December 1993; Accepted: 16 January 1994; Received at publishers: 14 March 1994

SUMMARY

Here we describe small mobilizable vectors based on the *Escherichia coli* plasmids pK18 and pK19. We combined the useful properties of the pK plasmids (e.g., multiple cloning site, *lacZ* α fragment, sequencing with M13 primers) with the broad-host-range transfer machinery of plasmid RP4 and a modified *sacB* gene from *Bacillus subtilis*. The new pK derivatives can be transferred by RP4-mediated conjugation into a wide range of Gram[−] and Gram⁺ bacteria, and should facilitate gene disruption and allelic exchange by homologous recombination. As an application example, the generation of a defined deletion of the *hom-thrB* genes in the chromosome of the Gram⁺ bacterium *Corynebacterium glutamicum* is presented.

INTRODUCTION

Gene disruption and replacement techniques are well established tools for the genetic engineering of Gram[−]

Correspondence to: Dr. A. Schäfer, Department of Genetics, Faculty of Biology, P.O. Box 100131, 33501 Bielefeld, Germany. Tel. (49-521) 106-5604; Fax (49-521) 106-5626; e-mail: pili@biogene1.uni-bielefeld.de

Abbreviations: A, absorbance (1 cm); aa, amino acid(s); ATCC, American type culture collection; B., *Bacillus*; bp, base pair(s); bhr, broad host range; C., *Corynebacterium*; C.g., *C. glutamicum*; E., *Escherichia*; hom, gene encoding homoserine dehydrogenase; kb, kilobase(s) or 1000 bp; Km, kanamycin; *lacZ* α , *lac* fragment for α -complementation; LB, Luria-Bertani (broth); LBG, LB containing 2 mg glucose/ml; MCS, multiple cloning site; mob, DNA region for mobilization by RP4; nt, nucleotide(s); Nx, nalidixic acid; *oriT*, origin of transfer replication; *oriV*, origin for vegetative replication; p, plasmid; PnA, Penassay (broth); PolIk, Klenow (large) fragment of *E. coli* DNA polymerase I; ^R, resistance/resistant; ^S, sensitive; *sacB*, gene encoding levansucrase; *thrB*, gene encoding homoserine kinase; Tn, transposon; wt, wild type.

bacteria. The manipulation of the genome of Gram⁺ strains, however, is often hampered by the lack of highly efficient transfer systems, since many strains are not readily accessible by transformation or electroporation procedures. Recent studies have shown that bacterial conjugation is a rather nonspecific process that probably works throughout the microbial world (Davies, 1990; Mazodier and Davies, 1991). Conjugation mediated by the broad-host-range (bhr) plasmid RP4 (Datta and Hedges, 1971) is suitable to transfer plasmid DNA from *E. coli* to a wide range of Gram[−] bacteria. Furthermore, RP4 is able to promote mobilization of plasmids from *E. coli* at partly high efficiency into several Gram⁺ genera, like *Streptococcus*, *Bacillus*, *Staphylococcus*, *Mycobacterium*, *Streptomyces*, *Corynebacterium*, *Brevibacterium*, *Arthrobacter* or *Rhodococcus* (Trieu-Cuot et al., 1987; Mazodier et al., 1989; Schäfer et al., 1990).

To facilitate the genetic engineering of especially

Gram⁺ strains we have combined the *bhr* transfer machinery of RP4 with the two convenient cloning vehicles pK18 and pK19 (Pridmore, 1987). Like other derivatives of plasmid pBR322 (Bolivar et al., 1977) these plasmids are based on the replicon pMB1 (Sutcliffe, 1979) and their host range therefore is restricted to *E. coli* and closely related species of the genera *Salmonella* and *Serratia*. They carry the Tn5 Km^R gene, which is an excellent selection marker in many bacteria (Parke,

1990). In addition, we have equipped the mobilizable derivatives pK18*mob* and pK19*mob* with a genetically modified *sacB* gene (Selbitschka et al., 1993) which confers sucrose-sensitivity to Gram⁻ and some Gram⁺ species (Steinmetz et al., 1983; 1985; Gay et al., 1985; Simon et al., 1991; Jäger et al., 1992). We show that the resulting plasmids pK18*mobsacB* and pK19*mobsacB* are useful tools to force and to detect rare double cross-over events.

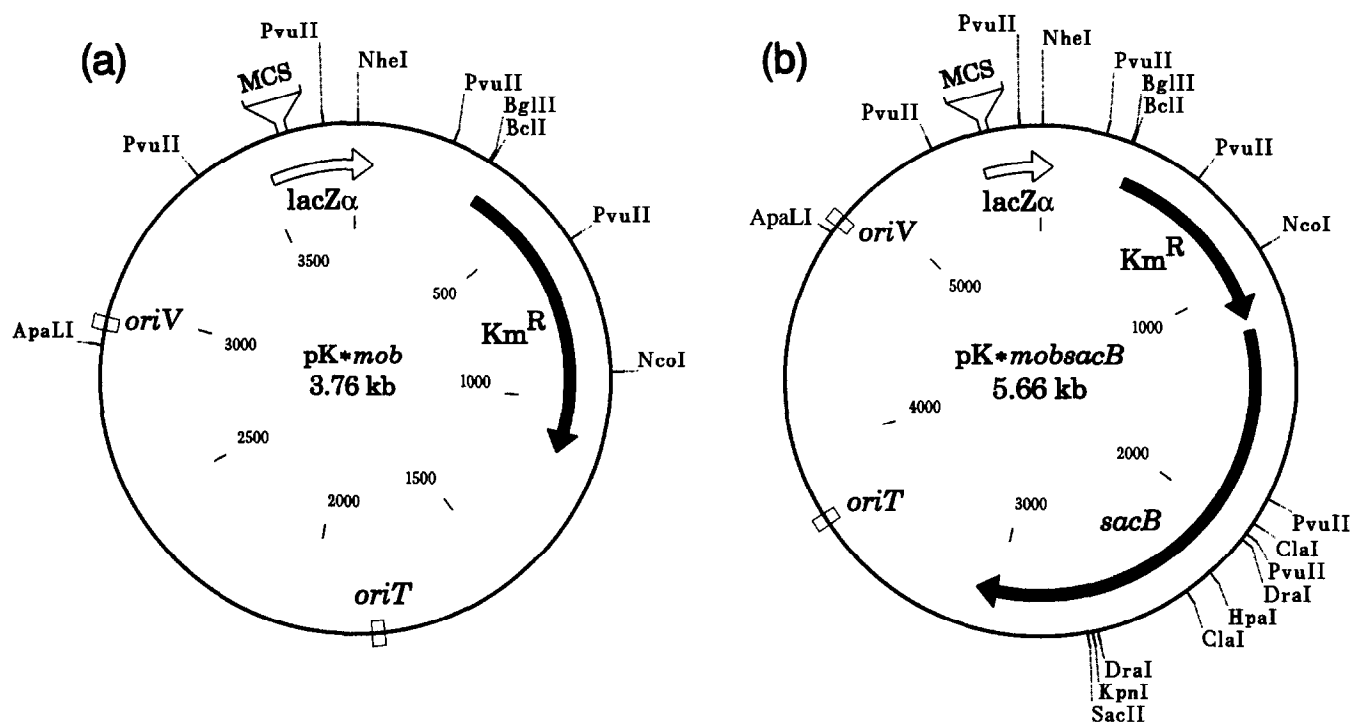


Fig. 1. Detailed physical maps of the mobilizable pK derivatives. The Km^R gene, the *lacZ* fragment and the *sacB* gene are represented by arrows indicating the direction of transcription. The *oriV* and the *oriT* (as a part of RP4*mob*) are depicted as small boxes. The asterisk symbolizes the plasmid labels 18 or 19, respectively. The enzyme recognition sites and the orientation of the MCS in the pK derivatives correspond to those in plasmids pK18 and pK19. Unique restriction sites for cloning purposes in the MCS of pK18*mob* and pK18*mobsacB* are: *EcoRI*, *SmaI*, *BamHI*, *XbaI*, *Sall*, *PstI*, *SphI* and *HindIII*. The universal primer-binding site is located downstream from the *HindIII* site and the reverse primer binds upstream the *EcoRI*-site of the MCS in the *lacZα* fragment. **Methods: (a) Construction of mobilizable pK18 and pK19 derivatives:** In order to clone the RP4 *mob* DNA region which is sufficient for mobilization by a co-resident RP4 helper plasmid, the pSUP102 plasmid (Simon et al., 1986) was digested with *AluI*. Plasmid pK18 was partially cleaved with *NaeI* and ligated to the *AluI* fragments of pSUP102. The ligation mixture was transformed (Hanahan, 1983) into *E. coli* S17-1 (Simon et al., 1983), a strain with an RP4 derivative integrated into the chromosome which provides the transfer functions necessary for mobilization. Transformants selected on PnA agar containing 50 µg Km/ml were pooled and grown in liquid LB medium to reach an $A_{580\text{ nm}}$ of 1. The cells were then mated with an equal number of *E. coli* MM 294 (Nx^R; Hanahan, 1983) recipient cells on cellulose nitrate filters (Sartorius, 2.5 cm diameter, 0.2 µm pore size). Transconjugants that arise from pK18 derivatives containing the RP4 *mob* DNA region were selected on LB agar containing 50 µg Km/ml and 50 µg Nx/ml. The plasmid which we designated pK18*mob* carries a 1.1-kb insertion and is efficiently mobilized by *E. coli* S17-1. To construct plasmid pK19*mob*, we digested pK18*mob* and pK19*mob* *ApaLI*+*BglII* and isolated the 2.5-kb *ApaLI*-*BglII* and 1.2-kb fragments, respectively, from an agarose gel. These fragments were ligated and after transformation to *E. coli* S17-1 the resulting plasmid pK19*mob* was tested by mating with *E. coli* MM 294 (Nx^R) for mobilization and by cleavage with *EcoRI* and *BglII* for the correct orientation of the MCS. **(b) Construction of pK18*mobsacB* and pK19*mobsacB*:** The *sacB* gene was obtained from plasmid pSM20 (Selbitschka et al., 1993). This plasmid contains a genetically modified *sacB* gene that lacks the *HindIII* and *EcoRI* restriction sites in the coding region. The corresponding sites in the MCS therefore are available for cloning purposes. Plasmid pSM20 was digested with *Ecl136II*+*XbaI* and the overhanging *XbaI* ends were filled in using PolIk. Plasmid pK18*mob* was cleaved with *AsuII* which generates a deletion of a 150-bp *AsuII* fragment from the plasmid. The remaining part of the vector was treated with PolIk and ligated to the 1.9-kb *SacB* fragment. The ligation mixture was transformed into *E. coli* DH5α (Hanahan, 1983) and selected on LB agar (50 µg Km/ml). Transformants containing pK18*mobsacB* were not able to grow on LB agar containing 10% sucrose. This demonstrates that the *sacB* gene is readily expressed to give levansucrase the action of which is lethal for cells growing in the presence of sucrose. Plasmid pK19*mobsacB* was obtained in the same way as pK18*mobsacB* but using pK18*mobsacB* instead of pK18*mob*. As pK18*mobsacB* plasmid pK19*mobsacB* is mobilizable and confers sucrose sensitivity to its *E. coli* host.

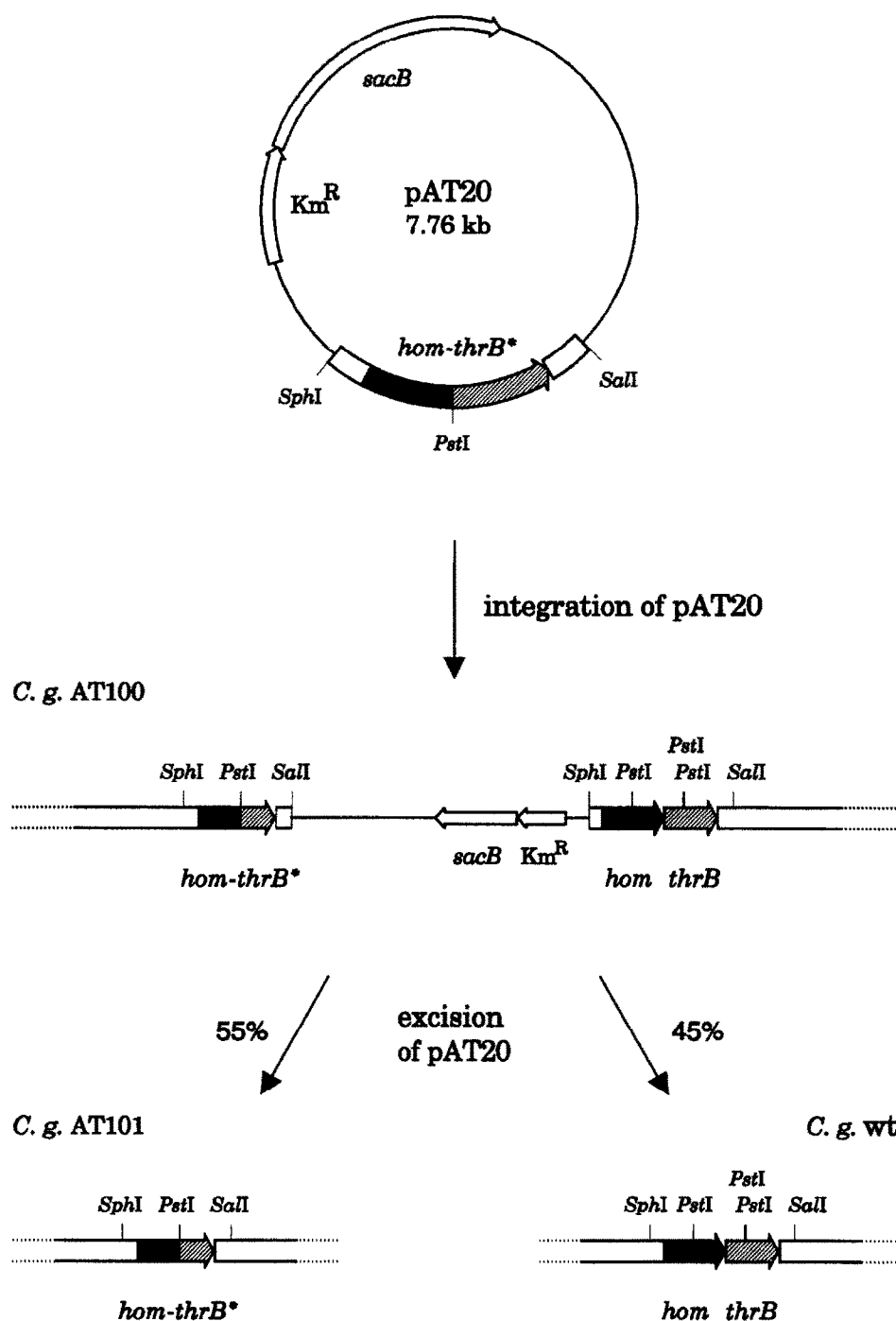


Fig. 2. Application of plasmid pK18mobsacB to generate a defined, marker-free deletion in the *hom-thrB* operon of *C. g.* The 3-kb *SphI-SalI* fragment of plasmid pJChom2 containing the *hom-thrB* operon was cloned into pK18mobsacB digested with *SphI* + *SalI*. The ligation mixture was transformed into *E. coli* DH5 α and transformants were selected on PnA agar containing 50 μ g *Km*/ml. The resulting plasmid pAT15 was cleaved with *PstI*, re-ligated and transformed into *E. coli* S17-1. The deletion derivative pAT20 carries a modified *hom-thrB* operon of *C. g.* (*hom-thrB**) that lacks the central 1-kb *PstI* fragment encoding the C-terminal 236 aa of *hom* and the 113 N-terminal aa of *thrB*. After mobilization of pAT20 into *C. g.*, the plasmid can only establish itself by integration into the chromosome via homologous recombination. The strain *C. g. AT100* carries the modified (*hom-thrB**) and the wt (*hom-thrB*) genes separated by vector sequences. Excision of the plasmid can be selected for by growing *C. g. AT100* cells on LBG agar containing 10% sucrose. Cells able to grow on this medium have lost the plasmid due to a second cross-over event that either restores the wild-type situation (45%) or leads to the mutant strain *C. g. AT101* (55%) which is auxotrophic for threonine and methionine. Please note that the frequency at which the desired mutants are obtained may vary depending on the length ratio of the homologous fragments adjacent to the deleted DNA region. Black boxes represent *hom* sequences and *thrB* sequences are depicted as hatched boxes. The DNA regions adjacent to the *hom thrB* genes are indicated as open boxes, the *Km^R* and *sacB* genes as open arrows. Vector sequences are symbolized by a single line. The asterisk marks the modified *hom-thrB* operon.

EXPERIMENTAL AND DISCUSSION

(a) Construction and advantages of mobilizable pK derivatives

Restriction maps of plasmids pK18*mob*, pK19*mob*, pK18*mobsacB* and pK19*mobsacB* are, shown in Fig. 1. The details of construction are given in the legend of Fig. 1. The plasmids pK18*mobsacB* and pK19*mobsacB* resemble vectors constructed for allelic exchange purposes in Gram⁻ bacteria by Kanina et al. (1991) and Selbitschka et al. (1993), but combine the following advantages: (i) they carry the Tn5 Km^R marker; (ii) they are small and offer a high cloning capacity; (iii) with the exception of *SacI* (there is a second *SacI* site in the *mob* region) and *KpnI* (a second site in *sacB*), the recognition sites of the MCS of the pK vectors remain unique; (iv) plasmids carrying insertions cloned into the MCS can be easily identified using the *lac* colour reaction; (v) inserts can be directly sequenced using the standard M13 sequencing primers.

(b) Applications for coryneform bacteria

We have exploited the plasmid pK18*mobsacB* for the genetic engineering of *Corynebacterium glutamicum*, a Gram⁺ strain with 53% G + C used mainly in the fermentative production of aa. For this strain and related microorganisms RP4-mediated intergeneric conjugation is the most efficient system to introduce foreign DNA (Schäfer et al., 1990). The genes encoding the enzymes homoserine dehydrogenase (*hom*) and homoserine kinase (*thrB*) involved in the biosynthesis of threonine and methionine constitute an operon in *C. glutamicum* ATCC 13032 (Peoples et al., 1988). This operon is present in the plasmid pJChom2 (B. Eikmanns, unpublished). A 3-kb *SphI*-*Sall* fragment of this plasmid containing the *hom-thrB* operon was cloned into pK18*mobsacB*. A central 1-kb *PstI* fragment comprising the C-terminal 236 aa of *hom* and the 113 N-terminal aa of *thrB* was deleted to give pAT20 (Fig. 2). Plasmid pAT20 was transferred by intergeneric conjugation from *E. coli* S17-1 to *C. glutamicum* ATCC 13032. Since pAT20 is unable to replicate in *C. glutamicum*, transconjugants only arise after integration of pAT20 into the chromosome by homologous recombination (Fig. 2). Transconjugants were selected on LBG agar + 25 µg Km/ml and 50 µg Nx/ml. The cells of this strain (*C. glutamicum* AT100) were prototrophic and not able to grow on LBG agar in the presence of 10% sucrose. To select double cross-over events a single colony was grown for 24 h in non-selective LBG medium at 30°C. About 2 × 10⁶ cells were plated onto LBG agar containing 10% sucrose and incubated for 48 h at 30°C. About 50% of the resulting colonies (102 out of 200) were sensitive to Km, indicating the excision of the plas-

mid by a second cross-over event. This second cross-over either restores the wt situation or leads to a mutant with the deletion left in the chromosome (Fig. 2). The 102 Km^S colonies were therefore tested for their ability to grow on minimal medium (Katsumata, 1984) and on minimal medium supplemented with threonine and/or methionine (1 mM each). Altogether, 56 colonies (55%) were auxotrophic and able to grow only in the presence of threonine and methionine. These bacteria contain the modified *hom-thrB* operon with the 1-kb *PstI* deletion. No plasmid sequences were left in the chromosome as proved by Southern blot analysis (data not shown). The strain generated by this procedure (*C. glutamicum* AT101) excreted L-lysine when cultivated under limited threonine supplementation (results not shown).

The application of the *sacB* system greatly facilitates the selection of rare double cross-over events in coryneform bacteria: a single colony of *C. glutamicum* cells containing pAT20 integrated in the chromosome was grown in non-selective LBG medium for 24 h at 30°C and subsequently plated onto LBG agar without sucrose. Out of 1000 colonies tested none had lost the plasmid in a second cross-over.

ACKNOWLEDGEMENTS

The authors would like to thank Dr. B. Eikmanns, KFA Jülich, for kindly providing pJChom2 and Dr. A. Schwarzer for critical comments on the manuscript. We thank S. Niemann for supply of plasmid pSM20. A. Schäfer acknowledges the receipt of a scholarship of the Studienstiftung des deutschen Volkes. This work was supported by BMFT grant 038409 in collaboration with Degussa AG.

REFERENCES

- Bolivar, F., Rodriguez, R.L., Greene, P.Y., Betlach, M.C., Heyneker, H.L., Boyer, H.W., Crosa, Y.H. and Falkow, S.: Construction and characterization of new cloning vehicles. II. A multipurpose cloning system. *Gene* 2 (1977) 95–113.
- Datta, N., Hedges, R.W., Shaw, E.J., Sykes, R.B. and Richmond, M.H.: Properties of an R-factor from *Pseudomonas aeruginosa*. *J. Bacteriol.* 108 (1971) 1244–1249.
- Davies, J.: Interspecific gene transfer: where next? *Trends Biotechnol.* 8 (1990) 198–203.
- Gay, P., LeCoq, D., Steinmetz, M., Berkemann, T. and Kado, C.I.: Positive selection procedure for entrapment of insertion sequence elements in Gram-negative bacteria. *J. Bacteriol.* 164 (1985) 918–921.
- Hanahan, D.: Studies on transformation of *Escherichia coli* with plasmids. *J. Mol. Biol.* 166 (1983) 557–580.
- Jäger, W., Schäfer, A., Pühler, A., Labes, G. and Wohlleben, W.: Expression of the *Bacillus subtilis sacB* gene leads to sucrose sensitiv-

- ity in the Gram-positive bacterium *Corynebacterium glutamicum* but not in *Streptomyces lividans*. J. Bacteriol. 174 (1992) 5462–5465.
- Kaniga, K., Delor, I. and Cornelis, G.R.: A wide-host-range suicide vector for improving reverse genetics in Gram-negative bacteria: inactivation of the *blaA* gene of *Yersinia enterocolitica*. Gene 109 (1991) 137–141.
- Katsumata, R., Ozaki, A., Oka, T. and Furuya, A.: Protoplast transformation of glutamate producing bacteria with plasmid-DNA. J. Bacteriol. 159 (1984) 306–311.
- Mazodier, P., Petter, R. and Thompson, C.: Intergeneric conjugation between *Escherichia coli* and *Streptomyces* species. J. Bacteriol. 171 (1989) 3583–3585.
- Mazodier, P. and Davies, J.: Gene transfer between distantly related bacteria. Annu. Rev. Genet. 25 (1991) 147–171.
- Parke, D.: Construction of mobilizable vectors derived from plasmids RP4, pUC18 and pUC19. Gene 93 (1990) 135–137.
- Peoples, O.P., Liebl, W., Bodis, M., Maeng, M., Follettie, M.T., Archer, J.A. and Sinskey, A.J.: Nucleotide sequence and fine structure analysis of the *Corynebacterium glutamicum* *hom-thrB* operon. Mol. Microbiol. 2 (1988) 53–62.
- Pridmore, R.D.: New and versatile cloning vectors with kanamycin-resistance marker. Gene 56 (1987) 309–312.
- Schäfer, A., Kalinowski, J., Simon, R., Seep-Feldhaus, A.H. and Pühler, A.: High frequency conjugal plasmid transfer from *Escherichia coli* to various Gram-positive coryneform bacteria. J. Bacteriol. 172 (1990) 1663–1666.
- Selbitschka, W., Niemann, S. and Pühler, A.: Construction of gene replacement vectors for Gram[−] bacteria using a genetically modified *sacRB* gene as a positive selection marker. Appl. Microbiol. Biotechnol. 38 (1993) 615–618.
- Simon, R., Priefer, U. and Pühler, A.: A broad host range mobilization system for *in vivo* genetic engineering: transposon mutagenesis in Gram-negative bacteria. Bio/Technology 1 (1983) 784–794.
- Simon, R., O'Connell, M., Labes, G. and Pühler, A.: Plasmid vectors for the genetic analysis and manipulation of rhizobia and other Gram-negative bacteria. Methods Enzymol. 118 (1986) 640–658.
- Simon, R., Hötte, B., Klauke, B. and Kosier, B.: Isolation and characterization of insertion sequence elements from Gram-negative bacteria by using new broad-host-range positive selection vectors. J. Bacteriol. 173 (1991) 1502–1508.
- Steinmetz, M., Le Coq, D., Ben Djemia, H. and Gay, P.: Analyse génétique de *sacB*, gène de structure d'une enzyme sécrétée, la lévansaccharase de *Bacillus subtilis* Marburg. Mol. Gen. Genet. 191 (1983) 138–144.
- Steinmetz, M., Le Coq, D., Aymerich, S., Gonzy-Treboul, G. and Gay, P.: The DNA sequence of the gene for the secreted *Bacillus subtilis* enzyme levansucrase and its genetic control sites. Mol. Gen. Genet. 193 (1985) 220–228.
- Sutcliffe, J.G.: Complete nucleotide sequence of the *Escherichia coli* plasmid pBR322. Cold Spring Harbor Symp. Quant. Biol. 43 (1979) 77–90.
- Trieu-Cuot, P., Carlier, C., Martin, P. and Courvalin, P.: Plasmid transfer by conjugation from *Escherichia coli* to Gram-positive bacteria. FEMS Microbiol. Lett. 48 (1987) 289–294.